DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078
(An Autonomous Institute affiliated to VTU, Approved by AICTE &ISO 9001: 2008 Certified)
Accredited by National Assessment & Accreditation Council (NAAC) with ‘A’ Grade

Department of Mechanical Engineering

Batch Number

37

Mini Project phase 1 Presentation

Title: DESIGN AND ANALYSES VTOL-TRICOPOTER

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Abstract

Aerial robotics is an emerging field combining both aerospace and robotics. Tricopter, an aerial robot, is an unmanned aerial vehicle which can be controlled from a remote location or can be made to fly autonomously based on pre-programmed flight plans using complex dynamic automation systems and which can be controlled manually.

The configuration of the Tricopter constructed is three arms spaced at 120 degrees separation to maintain continued stability even while hovering. Speed controllers are employed to control the speed of each motor. The three propellers connected to three DC brushless motors produce enough thrust so that the UAV takes-off to desired heights varying the throttle.

Literature survey

Design genetic algorithm optimization education software

- Waypoint tracking: The autopilot controllers tune autonomous flight waypoint tracking in preparation for advanced navigation research.

Controller for a Tricopter Fly Path Planning

- Dynamics of Motion: The triple tilting rotor UAVs, based on Newton-Euler mathematical formulation, which has three rotorcrafts speed and one tilt angle as four inputs, is presented with six Degree of Freedom (DOF).

Propeller Geometry

- size and pitch of the propeller used for the required thrust and manureability of UAV.

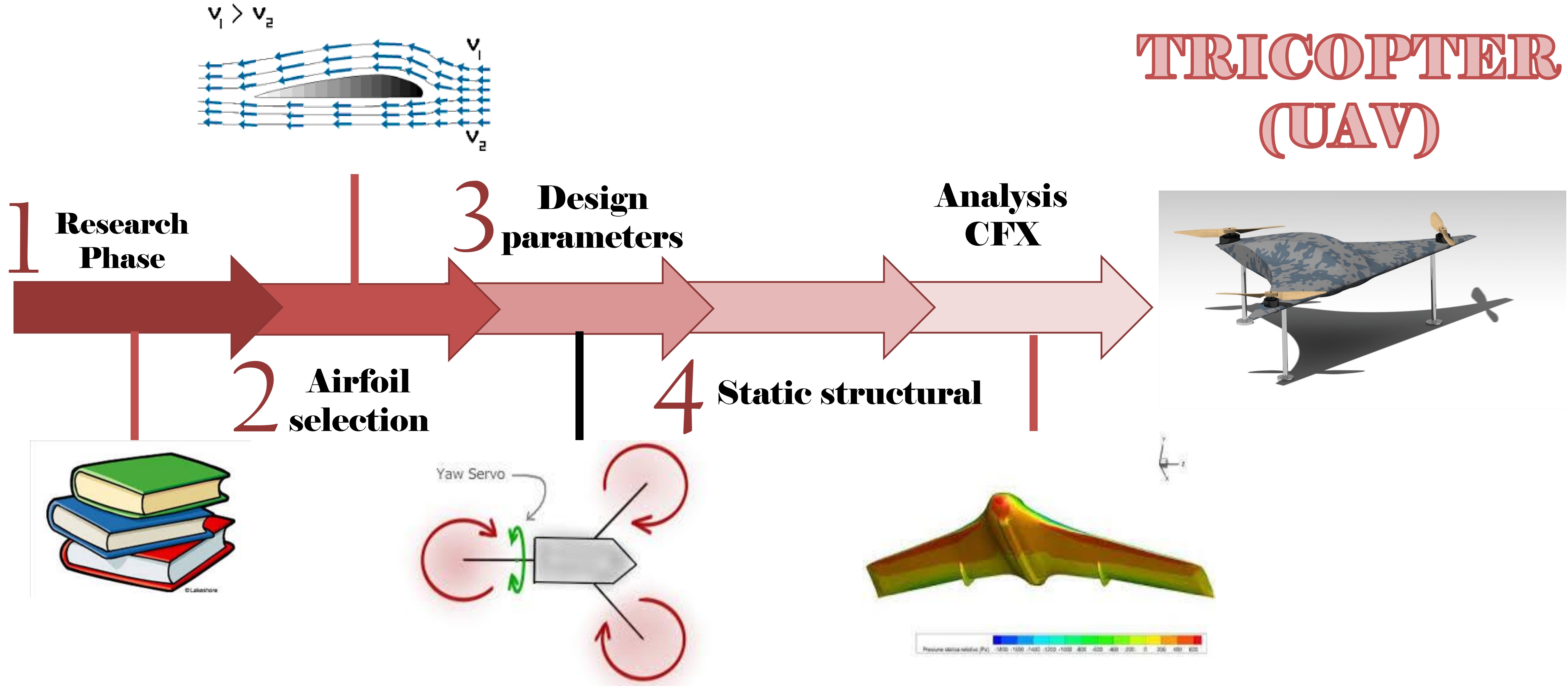
Aerodynamic design of wind turbine rotors

- The choice of airfoil series depends on the rotor design philosophy. It should be decided whether the blades should be slender or wide.

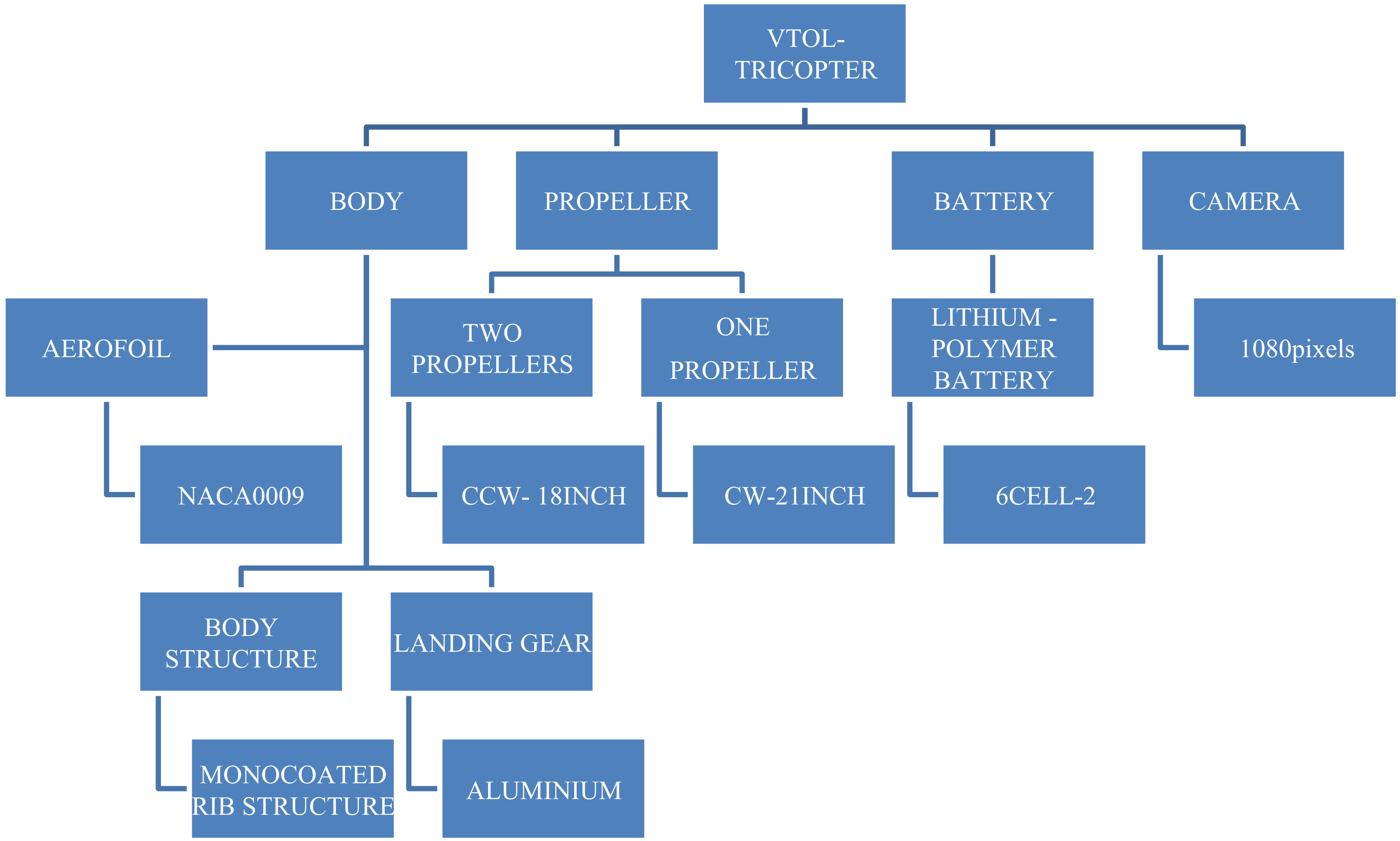
Objectives of the project:

- -To design a blended wing tri-copter.
- -To deign propellers for tri-copter.
- -To study the variation in deformation, stress and strain distributions .
- -To study computational fluid flow over the tri-copter, pressure counter and bending analysis for the UAV of specific dimensions.

Method and Methodology:



Design Constrains



CALCULATIONS

Propeller Calculation:-

$$F = 1.225\,\frac{\pi(0.0254\cdot d)^2}{4}\left[\left(RPM_p\cdot 0.0254\cdot p\cdot\frac{1}{60}\right)^2 - \left(RPM_p\cdot 0.0254\, p\cdot\frac{1}{60}\right)V_0\right]\left[\frac{d}{3.29546\cdot p}\right]^{1.}$$

- F → force obtained / required thrust
- d → diameter of propeller
- p → pitch of propeller
- V_0 → aircraft speed / Cruise speed
- RPM → rotations per minute

as our quadcopter weighs 2.5-3 kg & carries payload of 6-7 kg, then total weight will be around 10 kg.

∴ minimum required thrust will be 4-5 kg.

taking F = 5 kg ——— (i). η = RPM = 5000 (assumi

$$\Rightarrow 5 = 1.225\,\frac{\pi(0.0254\cdot d)}{4}\left[\left(5000\times 0.0254\times p\cdot\frac{1}{60}\right)^2 - \left(5000\times 0.0254\times p\cdot\frac{1}{60}\right)14\right]\left[\frac{d}{3.29546\,p}\right]^{1.5}$$

$$\Rightarrow \quad d/p = \frac{12}{8}.$$

∴ we got 12/8 so, we selected 12 × 8 propeller.

CALCULATIONS

chord dimension :-

Calculation:

w.k.t $R_e = \frac{\rho V D}{\mu}$ or $\frac{\rho V L}{\mu}$

in this case D can be replaced by L.

because D = L

$\mu = 1.81 \times 10^{-5}$ Kg/m· [Pa] → Constant

$\rho = 1.225$ Kg/m^3

$V = 10$ m/s

L = chord length = ?

R_e = 5,50,000 (constant)

$$\Rightarrow \quad 5{,}50{,}000 = \frac{1.225 \times 10 \times L}{1.81 \times 10^{-5}}$$

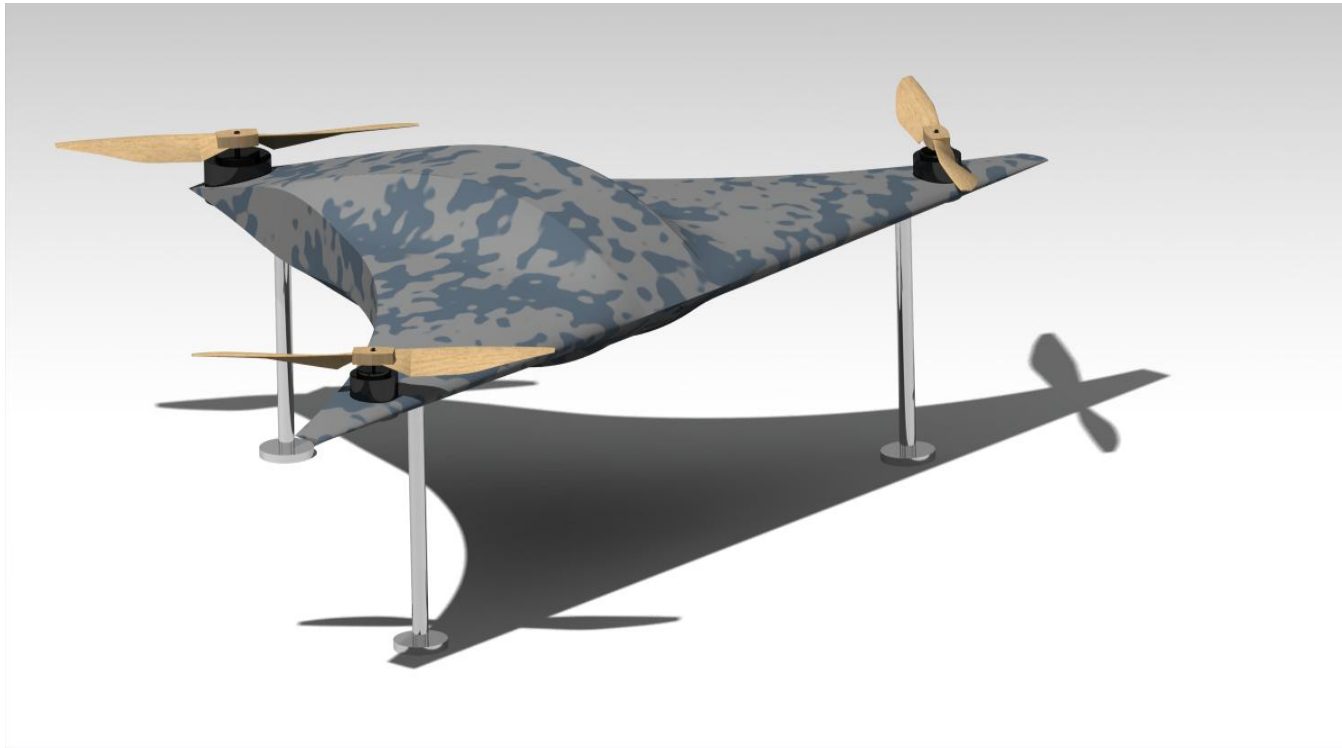
by calculating

$\Rightarrow$ L = 0.81 m

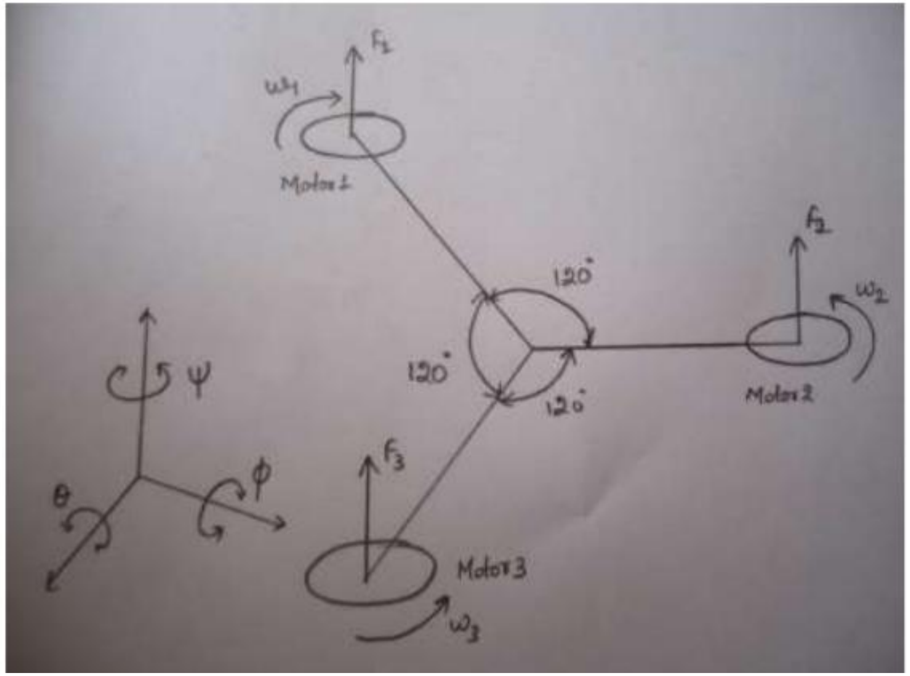
L ≃ 32 inches.

Design And Analyses

- ❖ **AIRFOIL:**NACA0009
- ▪ Root chord:32 in = 0.8128m
- ▪ Tip chord: 2 in = 0.0508m
- ❖ **BODY STRUCTURE**
- ▪ Blended wing configuration to obtain aerodynamic structure integrity.
- ▪ It is also house of power components and avionics parts.
- ❖ **PROPELLERS**
- ▪ FIALA Propellers
  Front: 12*8in =0.3048*0.2032m
  Rear: 14*10in =0.3556*0.254m
- ❖ **ANALYSES**
- ▪ Static structural.
- ▪ Computational Fluid Dynamics(CFX).



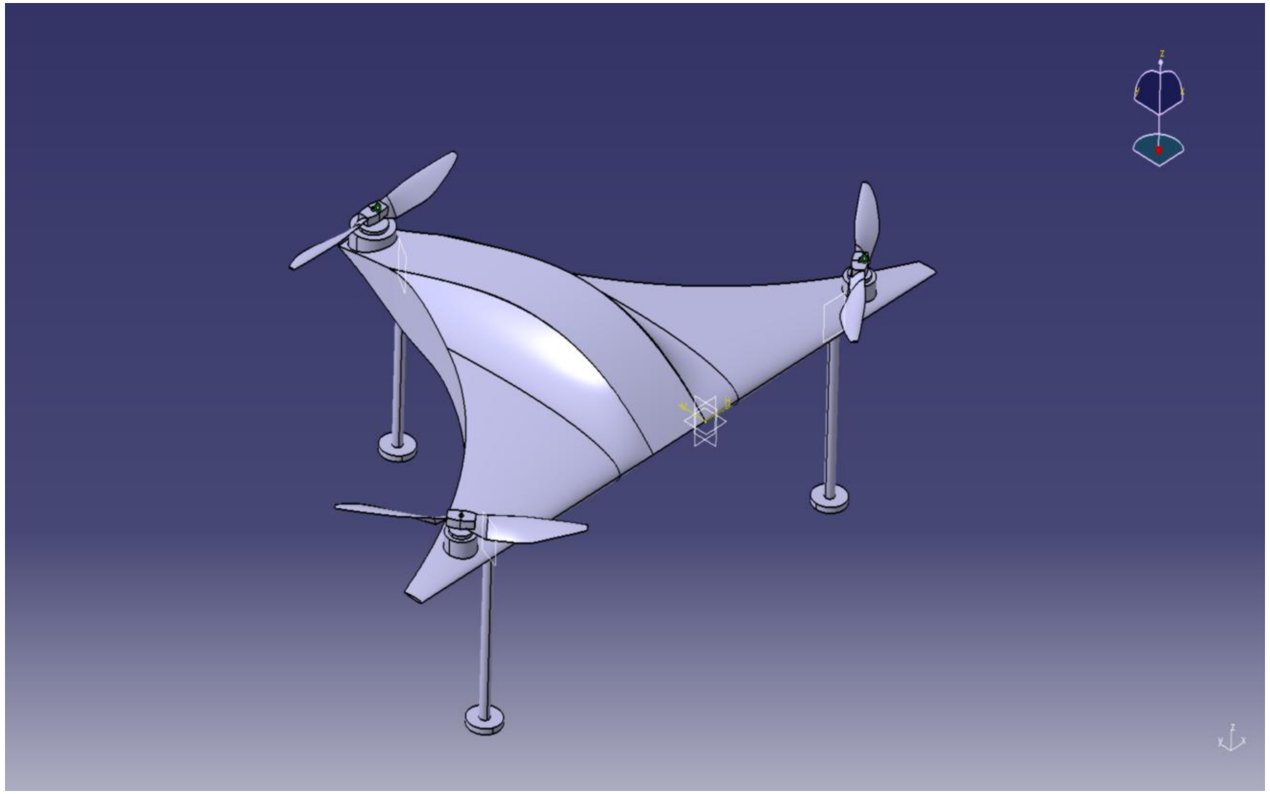
MOMENT OF INERTIA OF TRICOPTER



The dynamic equation of the tricopter in which the lower right side is the diagram of the dynamic equation of yaw control. That is because in yaw control, RC servomotor drives the tail axis to change the declination angle of the tail axis.

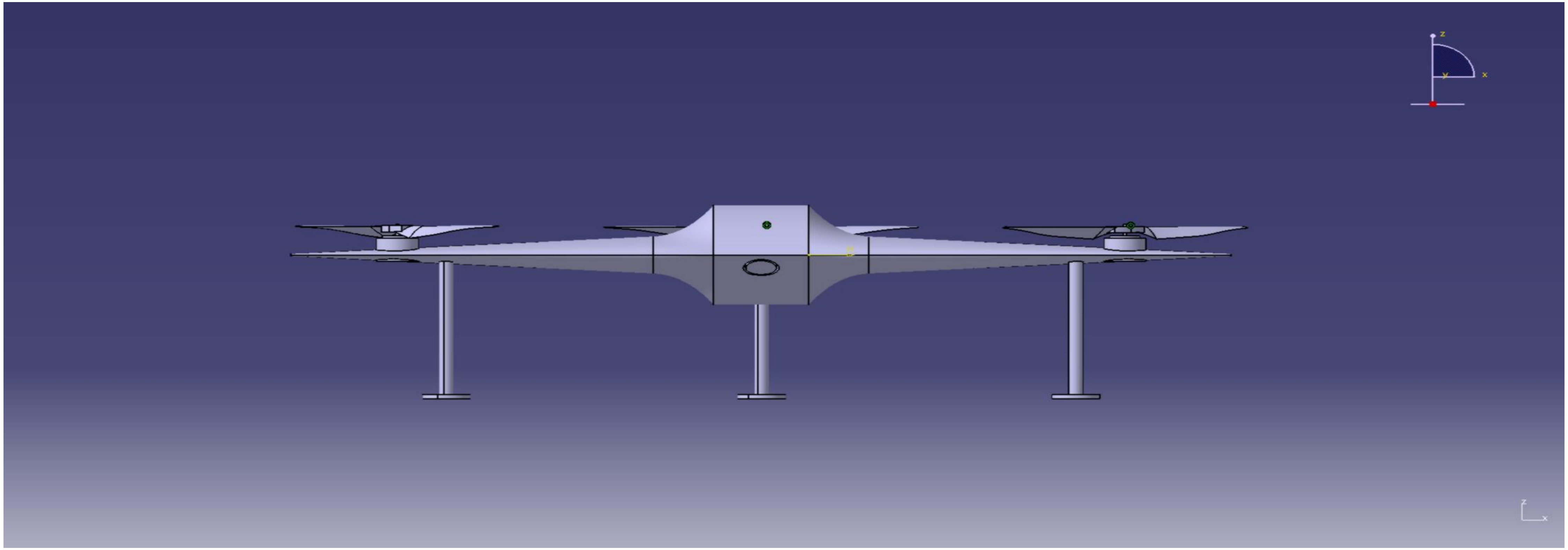
This is the most important to **balance the CG** of the tricopter.

DESIGN OF TRICOPTER



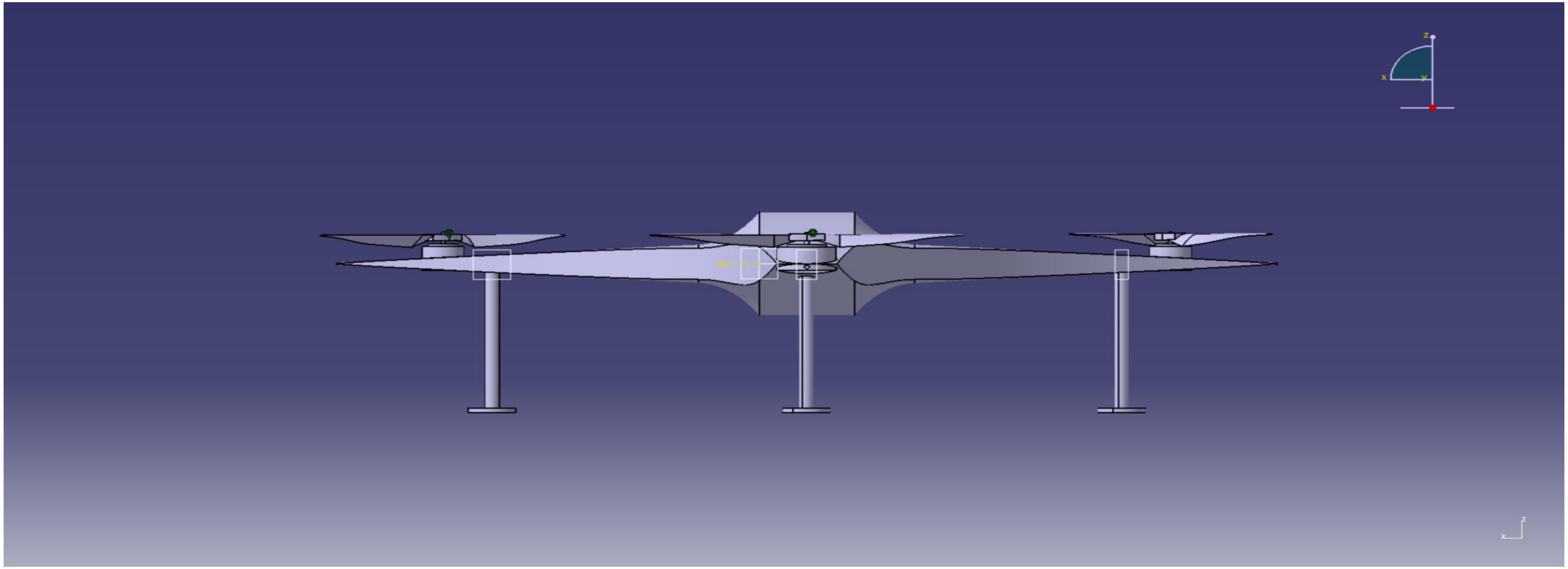
Sl.no	Description of side	measurements
1	Length on both ends of tricopter	2in = 0.0508m
2	Total inside thickness	15in = 0.318m
3	Propeller	12*8in =0.3048*0.2032m
4	Landing gear height	6in = 0.1524m
5	Motor dia	6in = 0.1524m

DESIGN VIEWS



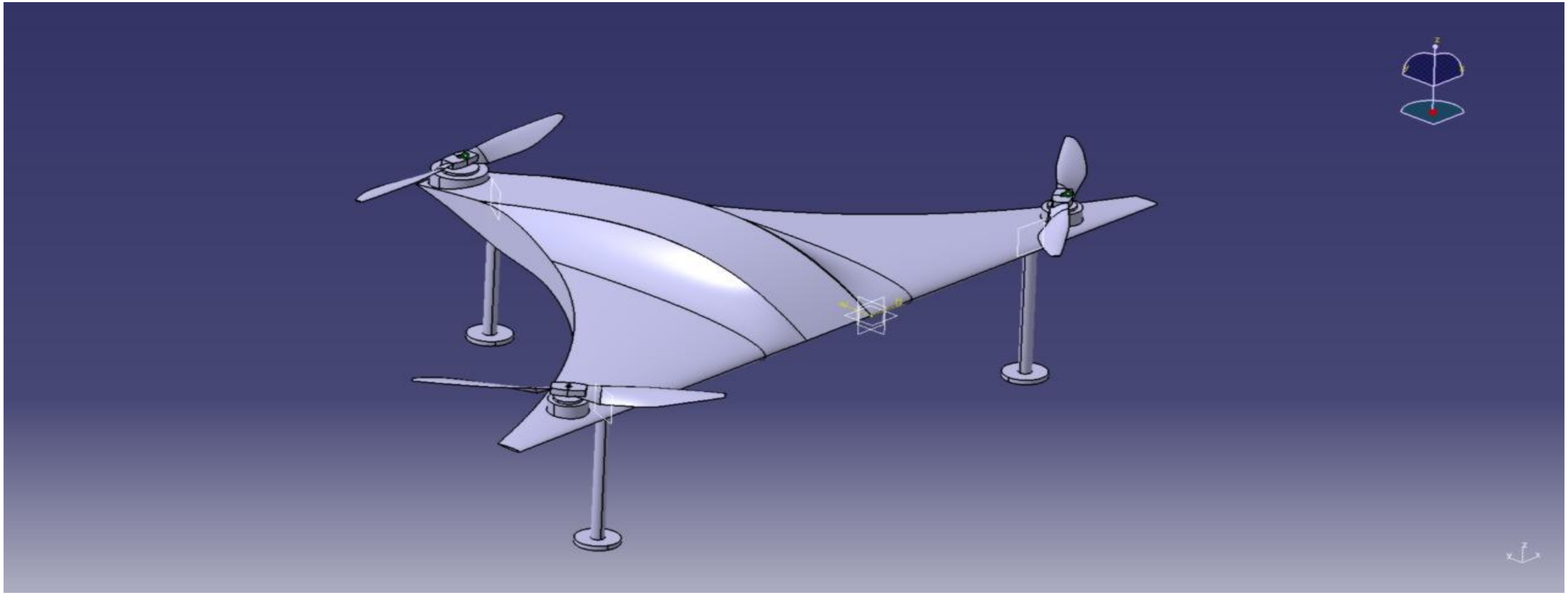
Front view

DESIGN VIEWS



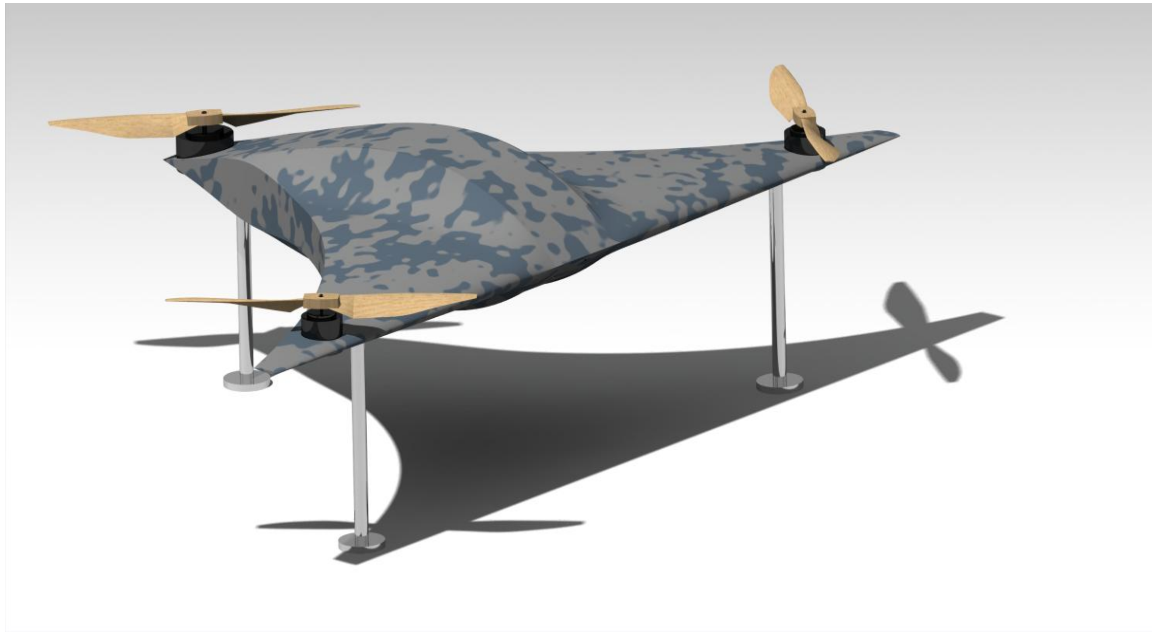
Rear view

DESIGN VIEWS



Iso view

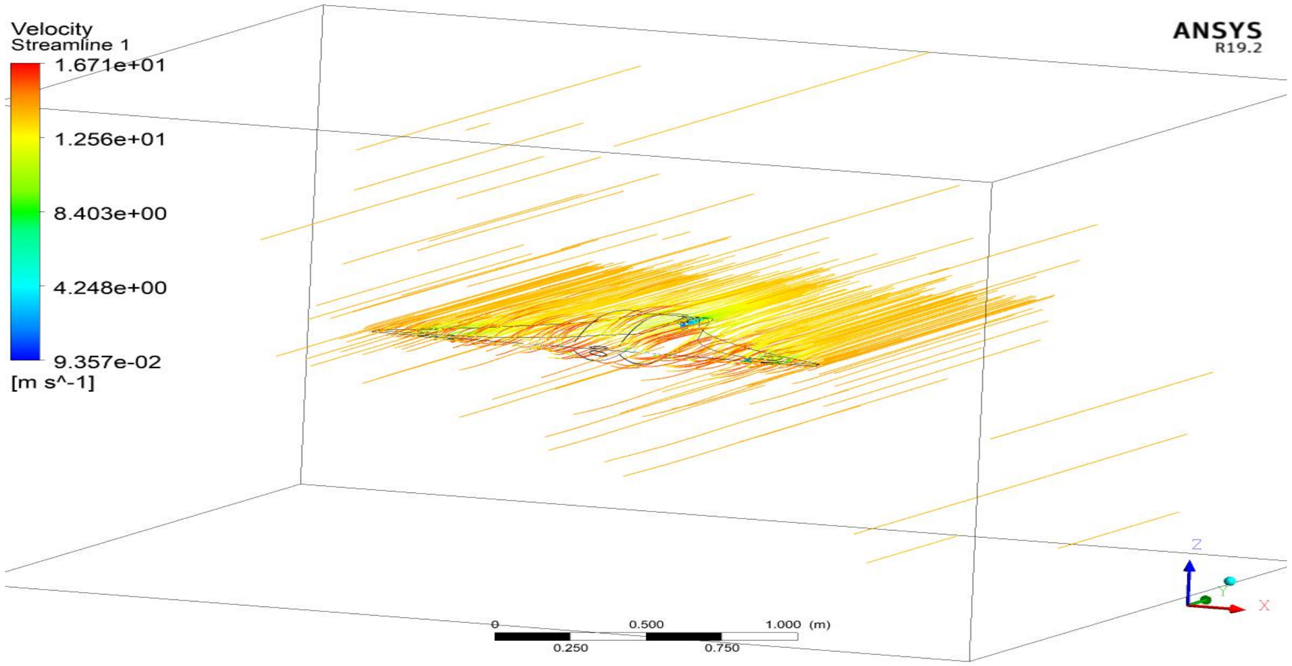
ANALYSIS OF TRICOPTER



- CFX-Computation Fluid Dynamics
- Static Structure
- Bending Analysis
- Tortional Test

	Constant	Comment
Reynolds Number(Re)	5,50,000	High
Wind Velocity	10m/s	Input
Angle Of Attack	0^0	Wind acts at 0^0 to the body
Load	100N	Max loading is 10kgs

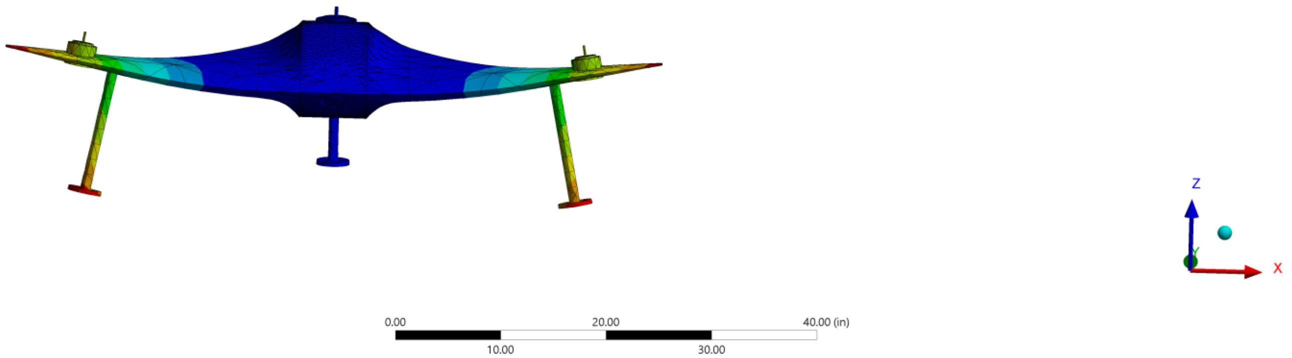
CFX of Tricopter



Here we have done cfx analysis of our design we have got negligible amount of drag value which is 4.5N,
which is very less and it shows that our design is absolutely perfect for the requirements.
Which also has streamlines in that picture and it depicts how the air flow occurs around the tricopter.

	Result	Comment
CFX AnalysisDrag Force	4.5N	less value shows that our design is absolutely perfect for the requirements

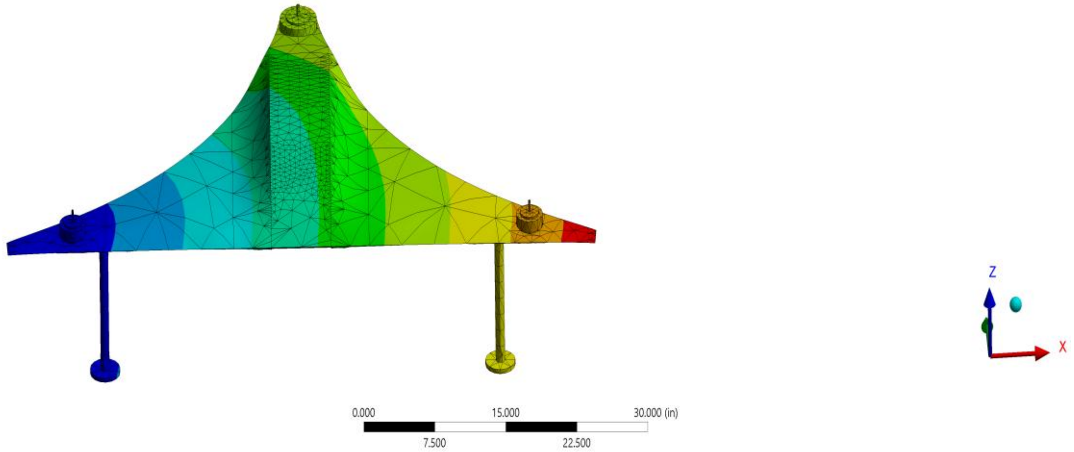
Bending Analysis on Tricopter



We have got 0.678*in* of deflection on each side of the deflection. Which is the maximum bending that can occur in our tricopter. The bending value which we got is very less and is states that our tricopter is stable during any bending condition even while carrying the maximum load.

	Result	Comment
Maximum Bending	0.678in=0.01722m of deflection on each side	very less and is states that our tri-copter is stable during any bending condition even while carrying the maximum load(10kgs)

Static Structure of Tricopter

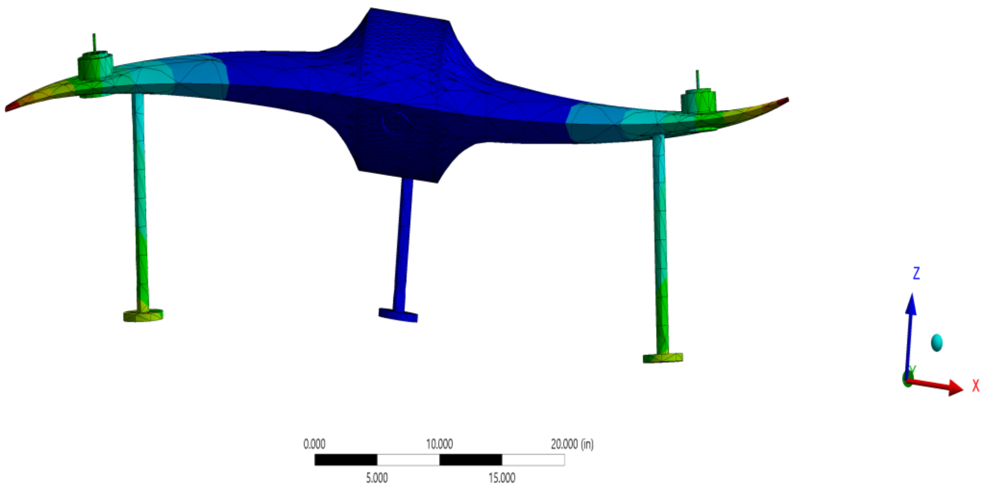


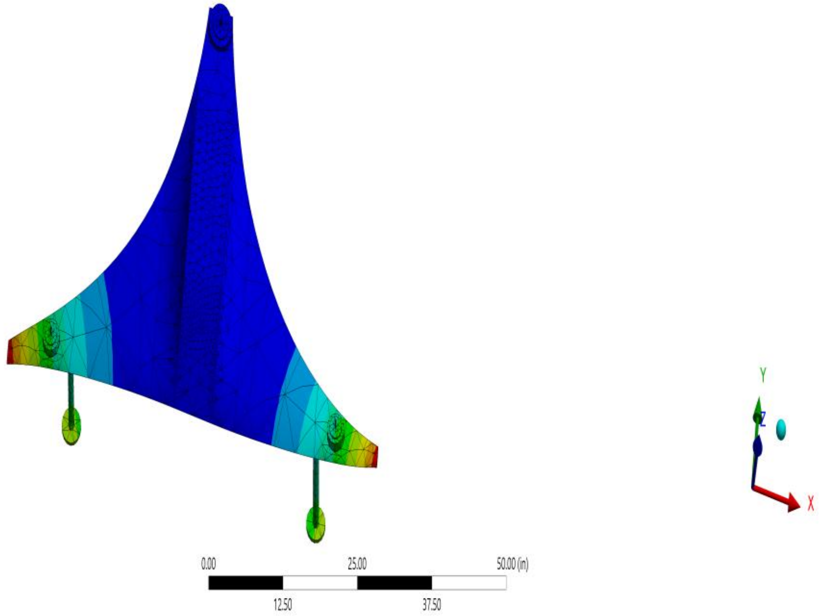
	Result	Comment
Static-Structural analysisMaximum Deformation Maximum Bending	0.3*in* =*0.00762m* of deflection on each side	very less magnitude of deformation and it shows that our aircraft is more stable in payload condition

We have done static structural analysis on our tricopter the obtained result is as shown

We have applied load of 100N because as we have mentioned earlier our tricopter can carry up to 10KG of load. the maximum deformation occurred is 0.3*in.* which is very less deformation and it shows that our aircraft is more stable in payload condition.

Deformations





Tortional test

We got the maximum deformation occurring in body is 0.5*in.=0.0127m*

Result

- Model of the tricopter with three propellers(used Naca0009 as airfoil shape).The virtual model had cubic enclosure as its boundary for the cfx analysis .
- Designed a blended wing tri-copter with propellers .
- -Studied the variation in deformation, stress and strain distributions .
- -Studied computational fluid flow over the tri-copter, pressure counter and bending analysis for the UAV of specific dimensions.